

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 08

DATE : 04-08-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 08

Topics

DATE : 04 - 08 - 2019

PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]

PHYSICS : Geometrical Optics, Electrostatics, Current Electricity, Capacitance, Wave Optics

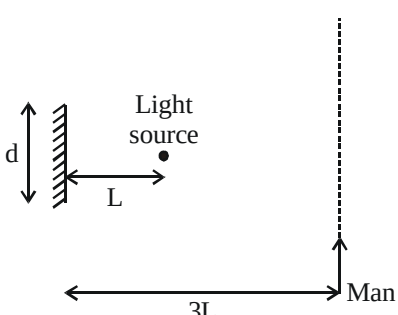
CHEMISTRY : Periodic Table, Chemical Bonding, GOC-I, IUPAC Nomenclature,
Co-ordination Compound

BIOLOGY : Genetics-I, Genetics-II [Upto Mutation], Human Health and Diseases

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

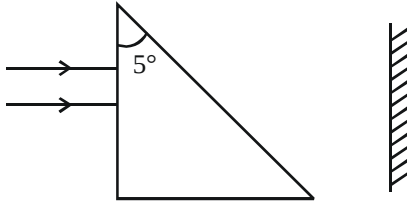
BEWARE OF NEGATIVE MARKING

- What should be the angle between two plane mirrors so that whatever be the angle of incidence, the incident ray and the reflected ray from the two mirrors be parallel to each other :-
 (1) 60° (2) 90°
 (3) 120° (4) 175°
- A plane mirror is approaching an object at a speed of 10 cm/sec. Calculate speed of image of object w.r.t. mirror.
 (1) 20 cm/sec (2) 10 cm/sec
 (3) 30 cm/sec (4) None of these
- A point source of light S is placed at a distance L in front of the centre of a mirror of width d, hangs vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance 3L from it as shown in figure. Find the distance over which he can see the image of light source.

- A ray of light falls on a glass slab of refractive index $\mu = \sqrt{3}$. What is the angle of incidence of the ray if the angle between the reflected and refracted ray is 90° .
 (1) 0° (2) 30°
 (3) 45° (4) 60°
- When the distance of an object from a concave mirror is decreased from 15 cm to 9 cm, the image gets magnified 3 times than that in first case. Calculate the focal length of the mirror.
 (1) 3 cm (2) 6 cm
 (3) 9 cm (4) 7.2 cm
- Which one of the following can produce a parallel beam of light from a point source of light ?
 (1) Concave mirror
 (2) Convex mirror
 (3) Concave lens
 (4) All of the above
- A bulb is placed at a depth of $2\sqrt{7}$ m in water and a floating opaque disc is placed over the surface of water so that the bulb is not visible from the surface. What is the minimum diameter of the disc.
 (1) 12 m (2) 6 m
 (3) 2 m (4) 4 m

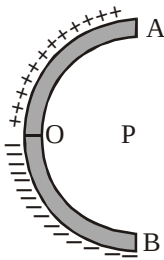
- (1) 3 d (2) 4 d
 (3) 2 d (4) d

SPACE FOR ROUGH WORK

8. A parallel beam of light is incident on a prism shown in figure. Through what angle should the mirror be rotated so that light returns back to its original path ? Refractive index of prism is 1.5.



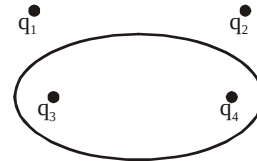
- (1) 5° (2) 2.5°
(3) 7.5° (4) None of these
9. A glass rod is bent into a semicircle of radius r . A charge $+Q$ is uniformly distributed along the upper half, and a charge $-Q$ is uniformly distributed along the lower half, as shown in figure. The electric field E at P , the center of the semicircle, is.



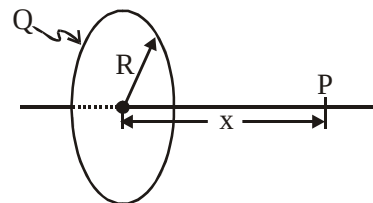
- (1) $\frac{Q}{\pi^2 \epsilon_0 r^2}$ (2) $\frac{2Q}{\pi^2 \epsilon_0 r^2}$
(3) $\frac{4Q}{\pi^2 \epsilon_0 r^2}$ (4) $\frac{Q}{4\pi^2 \epsilon_0 r^2}$
10. Two point charges repel each other with a force of 100 N. One of the charges is increased by 10%, and the other is reduced 10%. The new force of repulsion at the same distance would be.

- (1) 100 N (2) 121 N
(3) 99 N (4) None of these

11. Figure shows four charges q_1, q_2, q_3 and q_4 fixed in space. Then total flux of electric field through a closed surface S , due to all the charges, is



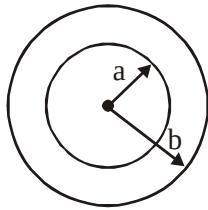
- (1) not equal to the total flux through S due to q_3 and q_4
(2) equal to the total flux through S due to q_3 and q_4
(3) zero if $q_1 + q_2 = q_3 + q_4$
(4) twice the total flux through S due to q_3 and q_4 if $q_1 + q_2 = q_3 + q_4$
12. A small particle of mass m and charge $-q$ is placed at point P on the axis of uniformly charged ring and released. If $R \gg x$, the particle will undergo oscillations along the axis of symmetry with an angular frequency that is equal to



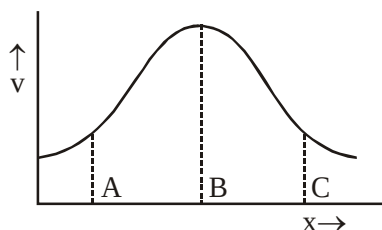
- (1) $\sqrt{\frac{qQ}{4\pi\epsilon_0 m R^3}}$ (2) $\sqrt{\frac{qQx}{4\pi\epsilon_0 m R^4}}$
(3) $\frac{qQ}{4\pi\epsilon_0 m R^3}$ (4) $\frac{qQx}{4\pi\epsilon_0 m R^4}$

SPACE FOR ROUGH WORK

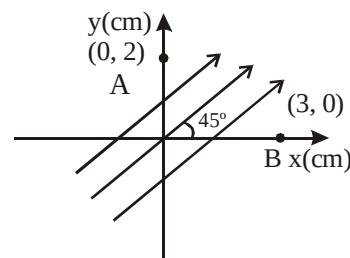
13. If the electric potential of the inner metal sphere is 10 volt & that of the outer shell is 5 volt, then the potential at the centre will be :



- (1) 10 volt (2) 5 volt
(3) 15 volt (4) 0 volt
14. 1000 identical drops of mercury are charged to a potential of 1 V each. They join to form a single drop. The potential of this drop will be:
- (1) 0.01 V (2) 0.1 V
(3) 10 V (4) 100 V
15. Let V be electric potential and E the magnitude of the electric field. At a given position, which of the statement is true ?
- (1) E is always zero where V is zero
(2) V is always zero where E is zero
(3) E can be zero where V is non zero
(4) E is always non zero where V is non zero
16. Variation of electrostatic potential along x -direction is shown in the graph. The correct statement about electric field is



- (1) x component at point B is maximum
(2) x component at point A is towards positive x -axis
(3) x component at point C is along negative x -axis
(4) x component at point C is along positive x -axis
17. Two identical charges are placed at the two corners of an equilateral triangle. The potential energy of the system is U . The work done in bringing an identical charge from infinity to the third vertex is.
- (1) U (2) $2U$
(3) $3U$ (4) $4U$
18. A uniform electric field of 400 V/m is directed at 45° above the x -axis as shown in the figure. The potential difference $V_A - V_B$ is given by



- (1) 0 (2) 4 V
(3) 6.4 V (4) 2.8 V
19. The capacity of a parallel plate condenser is $5\mu\text{F}$. When a glass plate is placed between the plates of the conductor, its potential becomes $1/8^{\text{th}}$ of the original value. The value of dielectric constant will be
- (1) 1.6 (2) 5
(3) 8 (4) 80

SPACE FOR ROUGH WORK

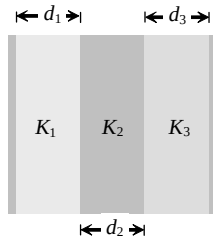
20. The expression for the capacity of the capacitor formed by compound dielectric placed between the plates of a parallel plate capacitor as shown in figure, will be (area of plate = A)

(1) $\frac{\epsilon_0 A}{\left(\frac{d_1}{K_1} + \frac{d_2}{K_2} + \frac{d_3}{K_3}\right)}$

(2) $\frac{\epsilon_0 A}{\left(\frac{d_1 + d_2 + d_3}{K_1 + K_2 + K_3}\right)}$

(3) $\frac{\epsilon_0 A(K_1 K_2 K_3)}{d_1 d_2 d_3}$

(4) $\epsilon_0 \left(\frac{AK_1}{d_1} + \frac{AK_2}{d_2} + \frac{AK_3}{d_3} \right)$



21. In a spherical condenser radius of the outer sphere is R. The difference in the radii of outer and inner sphere is x. Its capacity is proportional to

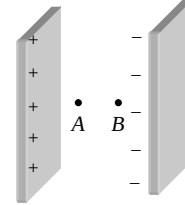
(1) $\frac{xR}{(R-x)}$

(2) $\frac{x(R-x)}{r}$

(3) $\frac{R(R-x)}{x}$

(4) $\frac{R}{x}$

22. Two protons A and B are placed in space between plates of a parallel plate capacitor charged upto V volts (See fig.) Forces on protons are F_A and F_B , then



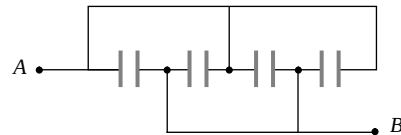
(1) $F_A > F_B$

(2) $F_A < F_B$

(3) $F_A = F_B$

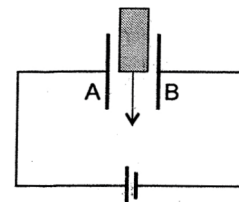
(4) Nothing can be said

23. Four condensers are joined as shown in the adjoining figure. The capacity of each is $8\mu F$. The equivalent capacity between the points A and B will be



- (1) $32\mu F$ (2) $2\mu F$ (3) $8\mu F$ (4) $16\mu F$

24. An insulator plate is passed between the plates of a capacitor. Then current:



(1) always flows from A to B

(2) always flows from B to A

(3) first flows from A to B and then from B to A

(4) first flows from B to A and then from A to B

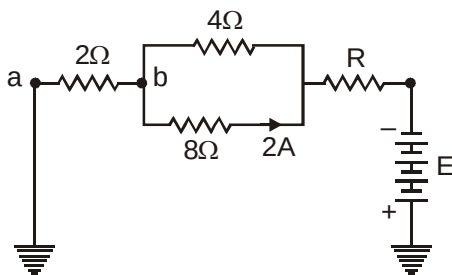
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25. The plate separation in a parallel plate condenser is d and plate area is A . If it is charged to V volt then the work done in increasing the plate separation to $2d$ will be –

(1) $\frac{3\varepsilon_0 AV^2}{2d}$ (2) $\frac{\varepsilon_0 AV^2}{d}$
(3) $\frac{2\varepsilon_0 AV^2}{d}$ (4) $\frac{\varepsilon_0 AV^2}{2d}$

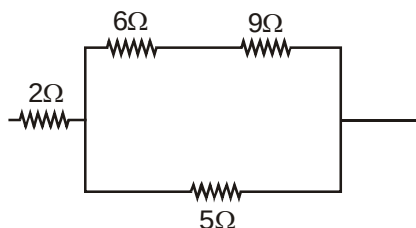
26. A parallel plate capacitor has an electric field of 10^5 V/m between the plates. If the charge on the capacitor plate is $1 \mu\text{C}$, the force on each capacitor plate is:
(1) 0.5 N (2) 0.05 N
(3) 0.005 N (4) none of these

27. In the circuit shown, the current in 2Ω resistor is



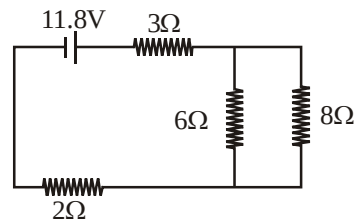
- (1) 2 A (2) 4 A (3) 6 A (4) 8 A

28. In the circuit shown in the figure the 5Ω resistor develops heat at the rate of 13.5 cal/s . The heat developed per second in the 2Ω resistor is



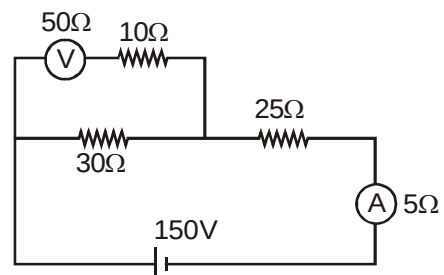
- (1) 1.2 Cal (2) 4.8 Cal
(3) 9.6 Cal (4) 10.8 Cal

29. In the network, the current in the resistor 6Ω is



- (1) $\frac{7}{5}$ A (2) $\frac{4}{5}$ A
(3) $\frac{3}{5}$ A (4) zero

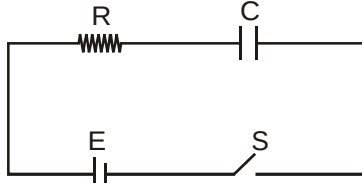
30. For given circuit, choose the correct option(s):



- (1) Reading of voltmeter is 70 V
(2) Reading of voltmeter is 100 V
(3) Reading of ammeter is 3A
(4) Reading of ammeter is 4A

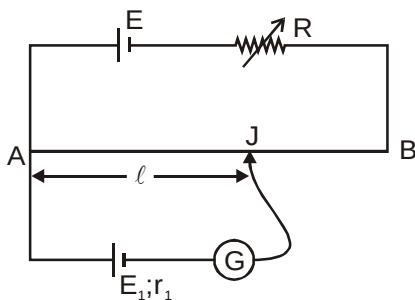
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31. At $t = 0$ switch S is closed. Find ratio of voltage across capacitor and voltage across resistance at time $t = 2RC \ln 3$ sec. (Capacitor is uncharged before switch S is closed).



- (1) 2 (2) 3 (3) 8 (4) 4

32. In potentiometer experiment, balance length (ℓ) is x and $2x$ when variable resistor R is set at value 10Ω and 24Ω . Resistance of potentiometer wire AB is (in Ω)

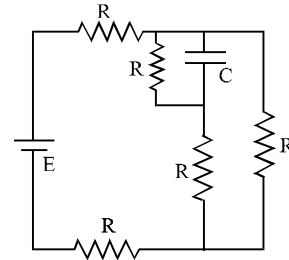


- (1) 10Ω (2) 8Ω (3) 4Ω (4) 20Ω

33. Two identical bulbs, when connected in series to a source, take $50W$. The total power consumed by them when connected in parallel with the same source is

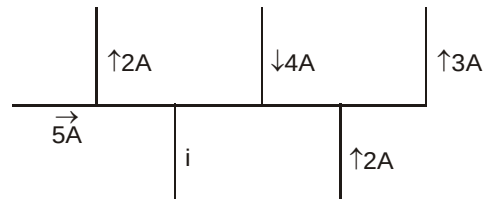
- (1) $25 W$ (2) $50 W$
(3) $100 W$ (4) $200 W$

34. Find the time constant of the circuit shown in figure.



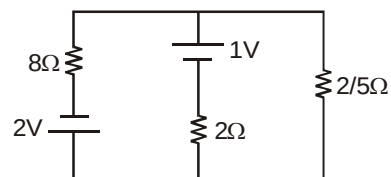
- (1) $2RC/3$ (2) $5RC/3$
(3) $5RC/8$ (4) none of these

35. A portion of a circuit is shown, with the values of the currents given for some branches. What is the direction and value of the current i ?



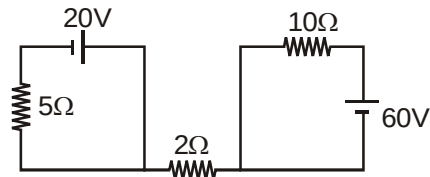
- (1) $\downarrow 6A$ (2) $\uparrow 6A$
(3) $\downarrow 4A$ (4) $\uparrow 4A$

36. The current flowing through $\frac{2}{5} \Omega$ resistor will be :-



- (1) $\frac{1}{5} A$ (2) $1A$ (3) $5A$ (4) $\frac{2}{5} A$

37. In the figure shown the current through 5Ω resistor is



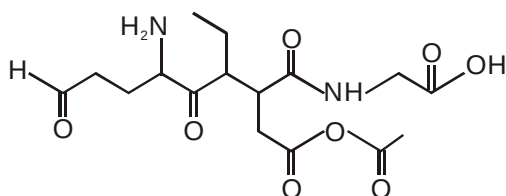
- (1) $2 A$ (2) $0 A$ (3) $4 A$ (4) $6 A$

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- | | |
|---|--|
| <p>38. The angle of polarisation for any medium is 60°. What will be critical angle for this?</p> <p>(1) $\sin^{-1}(\sqrt{3})$ (2) $\tan^{-1}(\sqrt{3})$ (3)</p> <p>$\cos^{-1}(\sqrt{3})$ (4) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$</p> <p>39. Two polarizers are oriented with transmission planes making an angle of 60°. The percentage of incident unpolarized light that passes through the system is</p> <p>(1) 50% (2) 100%</p> <p>(3) 12.5% (4) 37.5%</p> <p>40. In Young's experiment, the separation between 5th maxima and 3rd minima is how many times as that of fringe width?</p> <p>(1) 5 times (2) 3 times</p> <p>(3) 2.5 times (4) 2 times</p> <p>41. It is found that when waves from two identical coherent sources superpose at a certain point, the resultant intensity is equal to intensity of one wave only. This means that the phase different between the two waves at that point is</p> <p>(1) Zero (2) $\pi/3$</p> <p>(3) $2\pi/3$ (4) π</p> <p>42. In Young's double slit experiment, when two light waves form third minimum, they have</p> <p>(1) Phase difference of 3π</p> <p>(2) Path difference of 3λ</p> <p>(3) Phase difference of $\frac{5\pi}{2}$</p> <p>(4) Path difference of $\frac{5\lambda}{2}$</p> | <p>43. In a Young's double slit experiment, the fringe width is found to be 0.4 mm. If the whole apparatus is immersed in water of refractive index $\frac{4}{3}$ without disturbing the geometrical arrangement, the new fringe width will be</p> <p>(1) 0.30 mm (2) 0.40 mm</p> <p>(3) 0.53 mm (4) $450 \mu\text{m}$</p> <p>44. Consider an YDSE that has different slits width, as a result, amplitudes of waves from two slits are A and 2A, respectively. If I_0 be the maximum intensity of the interference pattern, then intensity of the pattern at a point where phase difference between waves is ϕ, is :-</p> <p>(1) $I_0 \cos^2 \phi$ (2) $\frac{I_0}{3} \sin^2 \frac{\phi}{2}$</p> <p>(3) $\frac{I_0}{9} [5 + 4 \cos \phi]$ (4) $\frac{I_0}{9} [5 + 8 \cos \phi]$</p> <p>45. In a standard YDSE setup, two coherent sources of light of intensity ratio β produces interference pattern. $\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$ will be equal to ($I_{\max}$ & $I_{\min}$ are maximum and minimum intensities of the resultant wave)</p> <p>(1) $\frac{2\sqrt{\beta}}{1-\beta}$ (2) $\frac{2\sqrt{\beta}}{1+\beta}$</p> <p>(3) $\frac{2\beta}{1+\beta}$ (4) $\frac{\sqrt{\beta}}{1+\beta}$</p> |
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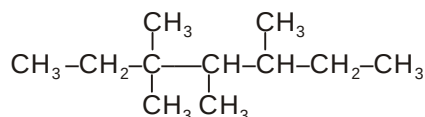
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46. The number of functional groups present in the following compounds is :



- (1) 5 (2) 7
(3) 6 (4) 8

47. In following compound



The correct lowest set of locant is :

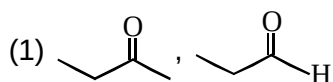
- (1) 3, 3, 4, 5
(2) 3, 4, 5, 5
(3) 4, 5, 3, 4
(4) 5, 5, 4, 3

48. The relation between given compound is :-

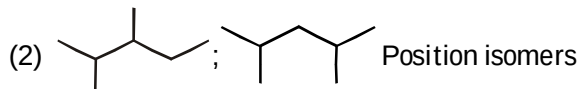


- (1) Chain isomerism
(2) Position isomerism
(3) Functional group isomerism
(4) Metamerism

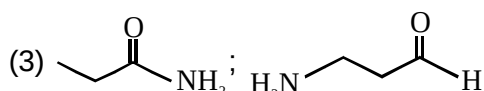
49. Which of the following is correct :-



Functional group isomers



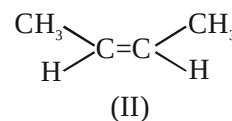
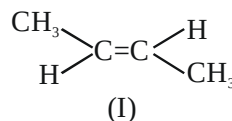
Position isomers



Position and functional group isomers

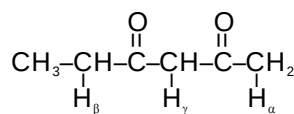
- (4) All of these

50. Which is true for the following isomeric forms I & II respectively :



Dipole moment	Boiling point	Melting point
(1) I > II	I > II	II > I
(2) II > I	II > I	II > I
(3) I > II	I > II	I > II
(4) II > I	II > I	I > II

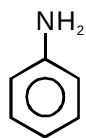
51. The order of acidic strength of the hydrogen atom (H_α , H_β , H_γ) in the given molecule is :



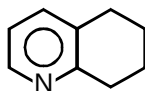
- (1) $H_\gamma > H_\alpha > H_\beta$ (2) $H_\gamma > H_\beta > H_\alpha$
(3) $H_\alpha > H_\beta > H_\gamma$ (4) $H_\beta > H_\alpha > H_\gamma$

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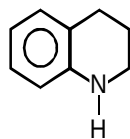
52. Arrange the following in decreasing order of basicity.



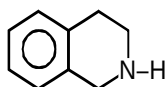
(I)



(II)



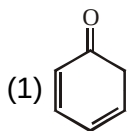
(III)



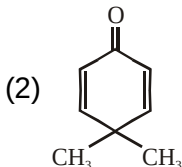
(IV)

- (1) I>III>IV>II
- (2) IV>I>III>II
- (3) IV>II>III>I
- (4) II<IV<III<I

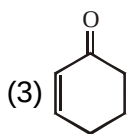
53. Which cannot show tautomerism ?



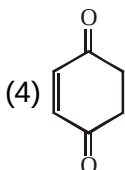
(1)



(2)

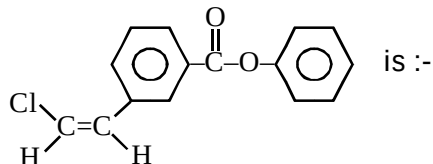
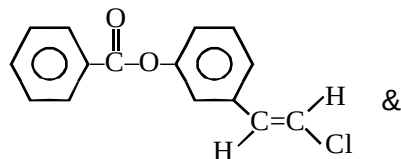


(3)



(4)

54. Isomerism shown by the following compounds



- (1) Functional group isomerism
- (2) Geometrical isomerism
- (3) Metamerism
- (4) Position isomerism

55. Match Column - I with Column - II and select the correct answer using the codes given below:

Column - I

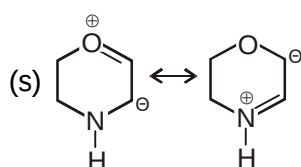
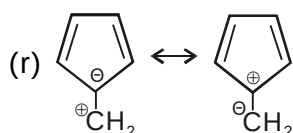
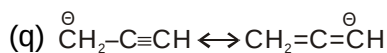
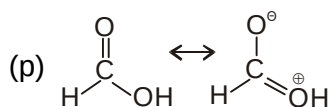
Column - II

- | | |
|---------------------------------|--|
| (A) Meso compound | (P) An equimolar mixture of enantiomers |
| (B) Enantiomers | (Q) Stereoisomers that are not mirror images |
| (C) Diastereomers | (R) Non superimposable mirror image |
| (D) Racemates (Racemic mixture) | (S) An optically inactive compound whose molecule is achiral even though they contain chiral centres |

- (1) A-S, B-R, C-Q, D-P
- (2) A-S, B-R, C-P, D-Q
- (3) A-R, B-S, C-Q, D-P
- (4) A-R, B-S, C-P, D-Q

SPACE FOR ROUGH WORK

56. For the following resonating structures



Identify more stable canonical structure

(1) (p) I > II (q) II > I (r) I > II (s) II > I

(1) (p) I < II (q) II > I (r) I > II (s) II > I

(1) (p) I < II (q) II > I (r) I < II (s) II > I

(1) (p) I > II (q) II < I (r) I > II (s) II < I

57. Which of the following statement is correct :-

(1) Number of e⁻s in pre penultimate shell of d-block elements are (1 – 10)

(2) 5th period contain 10 elements which has one or more than one 5d e⁻s.

(3) In modern periodic table there are 18 groups and 7 periods.

(4) Total no. of groups in p-block are 6 because p-orbital can hold maximum 6e⁻s.

58. Which of the following is not correctly matched

(1) [Xe] 4f¹⁴ 5d¹⁰ 6s² → Transition element

(2) [Rn] 5f¹⁴ 6d¹ 7s² → Inner transition element

(3) [Xe] 4f¹⁴ 5d¹⁰ 6s² 6p⁶ 7s² → Representative element

(4) [Xe] 4f¹⁴ 5d² 6s² → d-block element

59. Column –I
(Species)

Column –II
(Magnetic Moments)

(A) Ni⁺² (i) $\sqrt{35}$ B.M.

(B) O⁻² (ii) 0 B.M.

(C) Co⁺³ (iii) $\sqrt{8}$

(D) Fe⁺³ (iv) $\sqrt{24}$

(1) A–iii, B–ii, C–i, D–iv

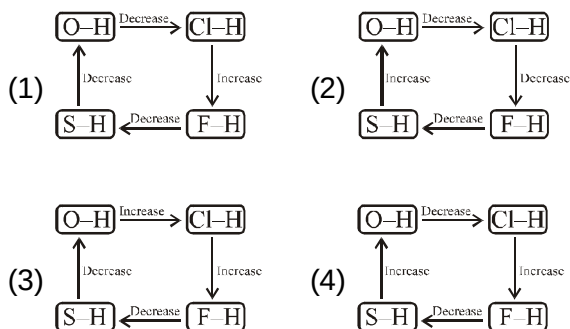
(2) A–ii, B–iii, C–iv, D–i

(3) A–iii, B–ii, C–iv, D–i

(4) A–iv, B–iii, C–ii, D–i

SPACE FOR ROUGH WORK

60. Which of the following diagrams shows correct change in the polarity of bond ?



61. An element has successive ionization enthalpies as 940 (first), 2080, 3090, 4140, 7030, 7870, 16000 and 19500 kJ mol⁻¹. To which group of the periodic table does this element belong ?

- (1) 14 (2) 15 (3) 16 (4) 17

62. Read the following statements:

(I) Atomic size of following pair are almost same

Fe, Co; Al, Ga; Zr, Hf

(II) E.N. order for the following will be

F > O > N > Cl

(III) Ionisation energy of

Al < Ga & Zr < Hf

(IV) Ionic radius

O²⁻ < F⁻ < Na⁺ < Mg⁺²

Select the correct statement(s)

- (1) I, III, IV
(2) I, II, III
(3) II, III, IV
(4) All the four

63. Which one of the following statements is incorrect in relation to ionization enthalpy?

- (1) Ionization enthalpy increases for each successive electron.
(2) The greatest increase in ionization enthalpy is experienced on removal of electron from core noble gas configuration.
(3) End of valence electrons is marked by a big jump in ionization enthalpy.
(4) Removal of electron from orbitals bearing lower n value is easier than from orbital having higher n value.

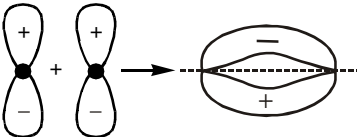
64. Which of the following statements is not correct from the point of view of molecular orbital theory ?

- (1) Be₂ is not a stable molecule
(2) He₂ is not stable but He₂⁺ is expected to exist
(3) Bond strength of N₂ is maximum amongst the homonuclear diatomic molecules
(4) The order of energies of molecular orbitals in F₂ molecule is $\pi_{2p_x} = \pi_{2p_y} < \sigma_{2p_z}$

65. The molecule which contain σ bonds, π bonds and lone pairs in 1 : 1 : 1 ratio :-

- (1) C₂N₂ (Cyanogen)
(2) C₆H₆ (Benzene)
(3) C(CN)₄ (Tetracyano methane)
(4) C₃O₂ (Carbon suboxide)

SPACE FOR ROUGH WORK

66. In CO_2 , SO_2 , SiO_2 each central atom is covalently bonded with :-
 (1) 2,2,4 oxygen atoms respectively
 (2) 2,2,2 oxygen atoms respectively
 (3) 2,4,4 oxygen atoms respectively
 (4) 4,4,4 oxygen atoms respectively
67. The incorrect order of bond angle :-
 (1) $\text{CO}_2 > \text{CO}_3^{2-} > \text{CF}_2\text{Cl}_2$
 (2) $\text{NO}_2^+ > \text{NO}_3^- > \text{NO}_2^-$
 (3) $\text{XeF}_2 > \text{XeO}_3 > \text{XeO}_4$
 (4) $\text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$
68. Which of the following combination of orbitals is correct ?
 (1) 
 (2) 
 (3) 
 (4) 
69. Which of the following set do not have sp^3d hybridization :-
 (1) PF_4^- & BrF_3 (2) ICl_2^+ & SF_4
 (3) XeF_2 & I_3^- (4) AsF_4^- & SCl_4
70. Ratio of σ & π bond in P_4O_{10} is :-
 (1) 4 (2) 3
 (3) 2 (4) 1
71. Which orbital is not involved in the formation of PCl_5 molecule :-
 (1) s (2) dz^2
 (3) dx^2-y^2 (4) p_z
72. Shape of molecule having 4-bond pair and one lone pair is :-
 (1) Trigonal bipyramidal
 (2) T-Shape
 (3) See-saw
 (4) Square planar
73. Which of the following order is incorrect :-
 (1) Ionic character = $\text{MCl} < \text{MCl}_2 < \text{MCl}_3$
 (2) Polarisability = $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$
 (3) Polarising power = $\text{Na}^+ < \text{Ca}^{+2} < \text{Mg}^{+2} < \text{Al}^{+3}$
 (4) Covalent character = $\text{LiF} < \text{LiCl} < \text{LiBr} < \text{LiI}$
74. Which of the following order is not correct :-
 (1) $\text{CrO} < \text{Cr}_2\text{O}_3 < \text{CrO}_3$
 (covalent character)
 (2) ZnO (Philosopher's wool)
 $> \text{ZnS}$ (Zinc blend) (ionic character)
 (3) $\text{BeCl}_2 > \text{MgCl}_2 > \text{CaCl}_2 > \text{SrCl}_2 > \text{BaCl}_2$
 (ionic character)
 (4) $\text{KCl} < \text{AgCl}$ (covalent character)
75. What is the charge on the complex $[\text{Cr}(\text{C}_2\text{O}_4)_2(\text{H}_2\text{O})_2]$ formed by $\text{Cr}(\text{III})$
 (1) +3 (2) +1
 (3) +2 (4) -1
76. In which of the following complex the nickel metal is in highest oxidation state?
 (1) $\text{Ni}(\text{CO})_4$
 (2) K_2NiF_6
 (3) $[\text{Ni}(\text{NH}_3)_6](\text{BF}_4)_2$
 (4) $\text{K}_4[\text{Ni}(\text{CN})_6]$

SPACE FOR ROUGH WORK

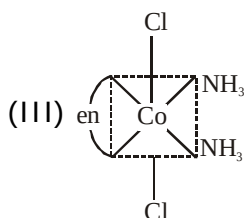
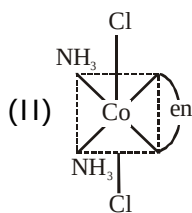
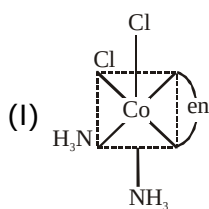
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| <p>77. How many moles of AgCl would be obtained, when 100 ml of 0.1M $\text{Co}(\text{NH}_3)_5\text{Cl}_3$ is treated with excess of AgNO_3?</p> <p>(1) 0.01 (2) 0.02
(3) 0.03 (4) none of these</p> <p>78. Among the following, metal carbonyls, the C-O bond is strongest in :</p> <p>(1) $[\text{Mn}(\text{CO})_6]^+$ (2) $[\text{Cr}(\text{CO})_6]$
(3) $[\text{V}(\text{CO})_6]^-$ (4) $[\text{Ti}(\text{CO})_6]^{2-}$</p> <p>79. Which of following pairs of name and formula of complexes, is not correct ?</p> <p>(1) Tetraamminecopper(II)sulphate : $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$
(2) Diamminesilver(I)chloride : $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$
(3) Potassiumhexacyanidoferrate(III) : $\text{K}_4[\text{Fe}(\text{CN})_6]$
(4) Potassiumamminepentachloridoplatinate(IV) : $\text{K}[\text{PtCl}_5(\text{NH}_3)]$</p> <p>80. In Tollen's reagent, The oxidation number coordination number and effective atomic number of central metal ion are respectively [Atomic number of Ag = 47]</p> <p>(1) +1, 2, 50 (2) +1, 2, 51
(3) +2, 1, 50 (4) +1, 1, 50</p> <p>81. One mole of the complex compound $\text{Co}(\text{NH}_3)_5\text{Cl}_3$ gives 3 moles of ion on dissolution in water. One mole of the same complex reacts with two moles of AgNO_3 solution to yield two moles of AgCl(s). The structure of the complex is—</p> <p>(1) $[\text{Co}(\text{NH}_3)_4\text{Cl}]\text{Cl}_2 \cdot \text{NH}_3$
(2) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
(3) $[\text{Co}(\text{NH}_3)_3\text{Cl}].2\text{NH}_3$
(4) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot \text{NH}_3$</p> | <p>82. Select the correct statement for the complexes $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ and $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$</p> <p>(1) There can be differentiated either by amount, nature or colour of PPT formed.
(2) There can be differentiated by electrical conducting measurement method.
(3) There can be differentiated by heating with conc. H_2SO_4.
(4) All of these</p> <p>83. The complexes $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_6]$ and $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2][\text{PtCl}_4]$ are :-</p> <p>(1) Linkage isomers
(2) Optical isomers
(3) Co-ordination isomers
(4) Ionisation isomers</p> <p>84. $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$ are related to each other as :-</p> <p>(1) Geometrical isomers
(2) Linkage isomers
(3) Coordination isomers
(4) ionisation isomers</p> <p>85. Consider the following statement and arrange in the order of true/false as given in the codes :-</p> <p>(a) $[\text{Cr}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ and $[\text{Cr}(\text{NH}_3)_4(\text{CN})_2][\text{Cr}(\text{NH}_3)_2(\text{CN})_4]$ are coordination isomers:-
(b) $[\text{Cr}(\text{Py})_2(\text{H}_2\text{O})_2\text{Cl}_2]\text{Cl}$ and $[\text{Cr}(\text{Py})_2(\text{H}_2\text{O})\text{Cl}_3]\text{H}_2\text{O}$ are hydrate isomer
(c) $[\text{Pt}(\text{NH}_3)_4\text{Br}_2]\text{Cl}_2$ and $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Br}_2$ are linkage isomers
(d) $[\text{NiCl}_2(\text{PPh}_3)_2]$ exhibits geometrical isomerism</p> <p>(1) TTFT (2) FTFT
(3) TTFE (4) TFFF</p> |
|--|---|

SPACE FOR ROUGH WORK

86. Which of the following is not correctly matched

- (1) d-block element : $(n-1)d^{1-10} ns^{0-2}$
- (2) p-block element : $ns^2 np^{1-6}$
- (3) s-block element : $(n-2)f^0 (n-1)d^0 ns^{1-2}$
- (4) f-block element : $(n-2)f^{1-14} (n-1)d^{1-10} ns^{1-2}$

87. Three arrangements are shown for the complex $[\text{Co(en)}(\text{NH}_3)_2\text{Cl}_2]^+$ pick up the correct statement:-



- (1) II and III are optical isomers
- (2) I & III are optical isomers
- (3) I & II are geometrical isomers
- (4) II & III are geometrical isomers

88. Select the correct statement from the following?

- (1) $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ both are colourless
- (2) $[\text{Co}(\text{NH}_3)_4\text{Br}_2]\text{Cl}$ show ionization isomers and geometrical isomers
- (3) $[\text{Pd}(\text{NO}_2)_2(\text{NH}_3)_2]$ is square planar and shows geometrical as well as linkage isomers
- (4) Both (2) and (3) are correct

89. In zeise's salt $\text{C}=\text{C}$ bond length is :-

Note $\left\{ \begin{array}{l} \text{C}-\text{C bond length in ethane} = 1.54 \text{ \AA} \\ \text{C}=\text{C bond length in ethene} = 1.34 \text{ \AA} \\ \text{C}\equiv\text{C bond length in ethyne} = 1.20 \text{ \AA} \end{array} \right\}$

- (1) 1.37 Å
- (2) 1.19 Å
- (3) 1.87 Å
- (4) 1.34 Å

90. Which amongst the following are organometallic compounds :-

- (i) $\text{Al}_2(\text{CH}_3)_6$ (ii) $\text{K}[\text{PtCl}_3(\text{C}_2\text{H}_2)]$ (iii) $\text{N}(\text{CH}_3)_3$
- (1) i only
- (2) iii only
- (3) i, ii, and iii
- (4) i and ii only

SPACE FOR ROUGH WORK

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| <p>91. Which character does not express in female :-
 (1) Hairy pinna
 (2) Colour blindness
 (3) Hypertrichosis
 (4) (1) & (3) Both</p> <p>92. How many type of gametes are formed in completely linked trihybrid plant :-
 (1) 8 (2) 16 (3) 4 (4) 2</p> <p>93. Which gene will show criss cross inheritance :-
 (1) Baldness
 (2) Mammary gland
 (3) Colourblindness
 (4) None of these</p> <p>94. How many linkage group present in mouse :-
 (1) 7 (2) 21
 (3) 13 (4) 23</p> <p>95. Distance between two linked gene in chromosome is measured in cM it is :-
 (1) Ratio of crossing over between them
 (2) Cross over value
 (3) Number of other gene between them
 (4) None of these</p> <p>96. One way of detecting X linked inheritance is :-
 (1) Son resembles mother and daughter resembles father
 (2) Son resembles father and daughter resembles mother
 (3) Both son and daughter resemble father
 (4) Both son and daughter resemble mother</p> <p>97. If human mother has O blood group her foetus blood can not be :-
 (1) A (2) B
 (3) AB (4) O</p> <p>98. If AB and O blood group are present in the offsprings of a couple which is true for this couple :-
 (1) Both are homozygous
 (2) Both are heterozygous
 (3) They have AB and O blood group
 (4) They can donate blood to any one</p> | <p>99. Colour blindness is found more in male than in female because :-
 (1) The males containing the single affected X-chromosome are colour blind
 (2) Heterozygous female are colour blind
 (3) Males having affected Y chromosomes are colour blind
 (4) All the above</p> <p>100. Which statement is not true for multiple allele :-
 (1) Multiple alleles are located on same locus of homologous chromosome
 (2) These are formed by mutation
 (3) A diploid organism has more than two alleles
 (4) Multiple alleles have different expression</p> <p>101. A trait is passed from man to only of his sons if.
 (1) Gene for that trait is dominant
 (2) Gene is located on Y chromosome
 (3) Both above
 (4) Located on X-chromosome</p> <p>102. During translation, peptide bond is formed between :-
 (1) NH_2 - group of P-site amino acid and - COOH group of A - site amino acid
 (2) - COOH group of P-site amino acid and -NH_2 group of A- site amino acid
 (3) -NH_2 group of P-site amino acid and -NH_2 group of A-site amino acid.
 (4) -COOH group of P-site amino acid and -COOH group of A-site amino acid.</p> <p>103. Translocase is an enzyme which is required for :-
 (1) DNA replication
 (2) RNA synthesis
 (3) Initiation of protein synthesis
 (4) Polypeptide chain elongation</p> |
|--|---|

104. Functional unit of gene that specifies one polypeptide chain is :-
 (1) Muton (2) Codon
 (3) Cistron (4) Recon
105. Usually triplet codons are translated in the direction :-
 (1) 3' → 5' (2) 3' → 3'
 (3) 5' → 3' (4) 5' → 5'
106. In prokaryotes ASD-sequence is present on which types of r-RNA :-
 (1) 18 's' rRNA (2) 16 's' rRNA
 (3) 23 's' rRNA (4) 5 's' rRNA
107. In protein synthesis first amino acid in polypeptide chain is normally methionine. But if methionine is not required in polypeptide chain then it is removed from polypeptide chain by :-
 (1) ATP
 (2) Peptidase enzyme
 (3) Amino peptidase enzyme
 (4) Peptidyle transferase enzyme
108. Consider the following :-
 (a) Structural gene
 (b) Messenger RNA
 (c) Ribosomes
 (d) Transcription
 (e) Translation
 Which one of the followings is the correct sequence for protein synthesis :-
 (1) a, d, b, c, e (2) a, d, c, b, e
 (3) a, b, d, c, e (4) c, e, d, b, a
109. The enzyme primase is a :-
 (1) DNA dependent DNA polymerase
 (2) RNA dependent DNA polymerase
 (3) DNA dependent RNA polymerase
 (4) RNA dependent RNA polymerase
110. The enzyme that does not synthesize nucleic acid is :-
 (1) Primase
 (2) Reverse transcriptase
 (3) RNA polymerase
 (4) Ligase
111. 'CCA' base sequence is present in which part of t-RNA :-
 (1) Anticodon loop
 (2) 3' end
 (3) 5' end
 (4) TΨC loop
112. In *E.coli rho-factor* is required for :-
 (1) Initiation of transcription at specific site
 (2) Initiation of transcription at many site
 (3) Termination of transcription of specific site
 (4) Termination of transcription at many site
113. During processing of Hn-RNA in eukaryotes after transcription, the following event is true :-
 (1) In tailing, adenylate residues are added at 5' end in a template dependent manner
 (2) The process of splicing represents the dominance of RNA world
 (3) The split gene arrangements represents probably an advanced feature of genome
 (4) In prokaryotes RNA polymerase II transcribe Hn-RNA
114. Consider the following four statements (a-d) and select the option which includes all the correct ones :-
 (a) The codon is read in mRNA in a contiguous fashion.
 (b) UAG codon has dual function in protein synthesis.
 (c) In actual structure, the tRNA is a compact molecule which looks like inverted 'L'
 (d) 28 's' rRNA in bacteria also behave as ribozyme.
 (1) b, c and d (2) a, b and d
 (3) a, c and d (4) a and c
115. Dominance is not an autonomous feature of a gene or the product, when :-
 (1) More than one gene influence the same phenotype.
 (2) More than one phenotype is influenced by the same gene.
 (3) More than one genotype is influenced by the same gene.
 (4) More than one allele are there for a character.

116. The length of DNA molecule greatly exceeds the dimensions of the nucleus in eukaryotic cells. How is this DNA accommodated?
- through elimination of repetitive DNA
 - deletion of non-essential genes
 - super-coiling in nucleosomes
 - DNase digestion
117. Haploids are more suitable for mutation studies than the diploids. This is because
- haploids are more abundant in nature than diploids
 - all mutations, whether dominant or recessive are expressed in haploids
 - haploids are reproductively more stable than diploids
 - mutagens penetrate in haploids more effectively than in diploids
118. Semiconservative replication of DNA was first demonstrated in :-
- Salmonella typhimurium*
 - Drosophila melanogaster*
 - Escherichia coli*
 - Streptococcus pneumonia*
119. Select the two correct statements out of the four (i – iv) given below about lac operon.
- Glucose or galactose may bind with the repressor and inactivate it
 - In the absence of lactose, the repressor binds with the operator region
 - The z-gene codes for permease
 - This was elucidated by Francis Jacob and Jacques Monod
- The correct statements are :-
- (ii) and (iii)
 - (i) and (iii)
 - (ii) and (iv)
 - (i) and (ii)
120. Which enzyme/s will be produced in a cell in which there is a non-sense mutation in the lac Y gene?
- β -galactosidase
 - Lactose permease
 - Transacetylase
 - Lactose permease and transacetylase
121. An abnormal human baby with "XXX" sex chromosomes was born due to
- fusion of two sperms and one ovum
 - formation of abnormal sperms in the father
 - formation of abnormal ova in the mother
 - fusion of two ova and one sperm
122. The movement of a gene from one linkage group to another is called
- Crossing over
 - Inversion
 - Duplication
 - Translocation
123. Which of the following biomolecules does have a phosphodiester bond?
- Nucleic acids in a nucleotide
 - Fatty acids in a diglyceride
 - Monosaccharides in a polysaccharide
 - Amino acids in a polypeptide
124. Which one of the following is not applicable to ss-RNA?
- Chargaff's rule
 - Complementary base pairing
 - 5' phosphoryl and 3' hydroxyl ends
 - Heterocyclic nitrogenous bases
125. In Lac operon system:
- Three structural genes are present in a sequence of z, y, a
 - Operator is present downstream to structural genes
 - Regulator gene functions as recognition site for RNA polymerase
- All are correct
 - Only (iii) is incorrect
 - Only (i) is correct
 - Only (ii) is correct

126. What is true for *E.coli* with inhibited lac-y-gene?

- (1) They cannot synthesize functional β -galactosidase
- (2) They cannot synthesize permease
- (3) They cannot transport lactose from medium into the cell
- (4) More than one option is correct

127. Some genes are constantly not expressing themselves in a cell and switched on or off according to the requirement of cellular activities, such genes are called

- (1) Non coding genes
- (2) House keeping genes
- (3) Luxury genes
- (4) Split genes

128. In Eukaryotes the gene regulation could be exerted at following levels:

- (i) Translational level
- (ii) m-RNA transport level
- (iii) Transcription level
- (iv) Processing level

The correct sequence of regulation is:

- (1) (iii) $\rightarrow$ (ii) $\rightarrow$ (i) $\rightarrow$ (iv)
- (2) (i) $\rightarrow$ (iv) $\rightarrow$ (iii) $\rightarrow$ (ii)
- (3) (iii) $\rightarrow$ (ii) $\rightarrow$ (iv) $\rightarrow$ (i)
- (4) (iii) $\rightarrow$ (iv) $\rightarrow$ (ii) $\rightarrow$ (i)

129. Select the option which correctly fills up the blanks in statements.

- (i) A is the substrate for the Enzyme β -galactosidase.
- (ii) B Provides the bindings site for repressor protein which prevent transcription of structural gene.
- (iii) C gene code for transacetylase Enzyme.

- (1) A - Lactose, B - Operator, C - Lac-a gene
- (2) A - Lactose, B - Regulator, C - Lac-a gene
- (3) A - glucose, B - Operator, C - Lac-Z gene
- (4) A - Allolactose, B - Operator, C - Lac-Z gene

130. Beadle and Tatum performed experiments on *Neurospora crassa* by creating mutants which were deficient in the production of specific substances. Such nutritional mutants are called:

- (1) Autotrophs
- (2) Auxotrophs
- (3) Prototrophs
- (4) Lethal

131. One gene one enzyme hypothesis was established for the first time in:

- (1) *Drosophila*
- (2) Pink bread mould
- (3) *E.coli*
- (4) *Pisum Sativum*

132. Peptidyl transferase activity to synthesize peptide bond between amino acids is:

- (1) 16 S-rRNA from smaller ribosomal unit
- (2) 5 S-rRNA of large subunit of ribosome
- (3) 28 S-rRNA of larger ribosomal subunit
- (4) 5.8 S-rRNA of larger subunit of ribosome

133. The UTRs are associated with (A) and present at (B)

	A	B
(1)	rRNA	5' end
(2)	mRNA	both 3' and 5' end
(3)	tRNA	3' end
(4)	mRNA	5' end only

134. Find out incorrect match.

- (1) Streptomycin-Inhibits initiation of translation
- (2) Erythromycin - Inhibits translocation of ribosome along m-RNA
- (3) Neomycin - Inhibits interaction between t-RNA and m-RNA
- (4) Chloramphenicol - Causes misreading

135. Severo Ochoa Enzyme is involved in

- (1) Polypeptide synthesis
- (2) DNA synthesis in template independent manner
- (3) RNA synthesis in template independent manner
- (4) Ribonucleotide polymerisation in template dependent manner

136. Which is not related with AIDS ?
 (1) It is a most fatal infectious disease
 (2) It is caused by HIV virus
 (3) NACO is working a lot to educate people about AIDS
 (4) Causative virus of AIDS is a single stranded DNA
137. Which of the following Statement is incorrect about interferon ?
 (1) It is species specific
 (2) It is useful in regression of Kaposi sarcoma
 (3) It is an example of physical barrier
 (4) It is secreted by virus infected cell
138. Which of the following is a secondary lymphoid organ ?
 (1) Thymus (2) Kidney
 (3) Bone marrow (4) Spleen
139. How many statements are correct about acquired immunity ?
 (a) Acquired immunity is a specific immunity
 (b) On second exposure of antigen this immunity shows anamnestic response.
 (c) B & T Lymphocytes are responsible for acquired immunity.
 (d) It can differentiate between self and non self
 (1) Two (2) Three (3) Four (4) One
140. Vaccination is based on :
 (1) Intensity of pathogen
 (2) Memory of person
 (3) Memory of immune system
 (4) Phagocytosis
141. Match the following columns and choose the incorrect matching :

	Column I	Column II
(1)	Spleen	Acts as a blood filter
(2)	Thymus	Size remains the same after birth
(3)	Phagocytosis	PMNL
(4)	Natural killer cells	Perforin

142. This is a diagram of ringworm of face, Which is not related with the disease?



- (1) *Microsporum*
 (2) *Epidermophyton*
 (3) *Wuchereria*
 (4) *Trichophyton*
143. Which of the following statement is incorrect about amoebic dysentery ?
 (1) It is caused by *Entamoeba histolytica*
 (2) Culex mosquito act as a mechanical carrier
 (3) It is a parasite of large intestine
 (4) Patients suffer from constipation, abdominal pain and cramps.
144. A patient consumed contaminated food few days before, now he is suffering from sustained high fever, headache, weakness, loss of appetite. What can be the probable disease?
 (1) Typhoid (2) Pneumonia
 (3) Diphtheria (4) Tuberculosis
145. The rupture of RBC is associated with the release of a toxin
 (1) Hirudin (2) Anophelin
 (3) Heparin (4) Haemozoin
146. How many of following diseases are vector borne?
 Malaria, Cholera, Swine flu, Dengue, Elephantiasis, AIDS
 (1) Two (2) Three
 (3) Four (4) Seven
147. Which type of bond is present in between light and heavy chain of an antibody?
 (1) Ionic bond (2) Peptide bond
 (3) Dioxide bond (4) Disulphide bond

148. Allograft is
- (1) Grafting between the individuals of different species
 - (2) Grafting between individuals of same species
 - (3) Heterograft
 - (4) Isograft
149. Which antibody indicate the presence of infection in new born baby?
- (1) IgM
 - (2) IgG
 - (3) IgA
 - (4) IgD
150. If a person is infected with some deadly microbes to which quick immune response is required as in tetanus, what treatment will be suggested to that patient ?
- (1) Recombinant DNA vaccine
 - (2) Passive immunization
 - (3) Active immunization
 - (4) hyposensatization
151. Which of the following are included in prophylaxis?
- a. Antiserum
 - b. Toxoid
 - c. Vaccines
 - d. Antivenom
 - e. Triple antigen (DPT,MMR)
- Choose the correct option from below
- (1) a, b, c, d, e
 - (2) b, c, e
 - (3) a, c, d, e
 - (4) a, b, c, d
152. Consider the following diseases
- a. Gonorrhoea
 - b. Anthrax
 - c. Ascariasis
 - d. Plague
 - e. Tetanus
- How many of the above are bacterial diseases?
- (1) Five
 - (2) Two
 - (3) Four
 - (4) One
153. Heart valve of pig may be used in human because
- (1) Same HLA
 - (2) Heavy dose of immunosuppressant is given
 - (3) Heart valve is vascular
 - (4) Heart valve is avascular
154. Type of antibody responsible for rheumatoid arthritis is ?
- (1) IgE
 - (2) IgM
 - (3) IgD
 - (4) IgA
155. Disease caused due to deficiency of enzyme adenosine deaminase is ?
- (1) AIDS
 - (2) SCID
 - (3) IDDM
 - (4) Hay fever
156. Allergy of lower respiratory tract is :-
- (1) Hay fever
 - (2) Pneumonia
 - (3) Asthma
 - (4) Myasthenia gravis
157. Intestinal perforation and death may occur in severe cases of which of the following ?
- (1) Pneumonia
 - (2) Typhoid
 - (3) Diphtheria
 - (4) Break-bone fever
158. Which of the following is not a viral disease ?
- (1) Mumps
 - (2) Swine flu
 - (3) Leprosy
 - (4) Rabies
159. Shick test is done for the detection of :-
- (1) Typhoid
 - (2) Hepatitis-B
 - (3) Plague
 - (4) Diphtheria
160. Which disease is spread by mosquito ?
- (1) Ascariasis
 - (2) Filariasis
 - (3) Black death
 - (4) Hepatitis-B
161. In malaria formation of gametocytes takes place in :-
- (1) Gut of mosquito
 - (2) Salivary gland of mosquito
 - (3) Liver of human
 - (4) RBC of human
162. Dry scaly lesions on various parts of body are seen in :-
- (1) Amoebiasis
 - (2) Filariasis
 - (3) Ringworms
 - (4) Chlamydiasis

163. Below are some statements related to life cycle of plasmodium. Which of the following is correct?

- (1) Gametocyte stage infects human body
- (2) It multiplies by sexual method of reproduction in human liver
- (3) human RBC are ruptured and releases a toxin hemozoin
- (4) development of gametocyte possible only in female culex mosquito

164. Which of the following is correct regarding AIDS causative agent HIV ?

- (1) HIV is unenveloped retrovirus.
- (2) HIV may or may not has a latent phase.
- (3) HIV is enveloped virus containing one molecule of single-stranded RNA and one molecule of reverse transcriptase.
- (4) HIV is enveloped virus that contains two identical molecules of single-stranded RNA and two molecules of reverse transcriptase.

165. Match each disease with its correct type of vaccine:

- | | |
|-------------------------|-------------------------|
| (a) tuberculosis | (i) harmless virus |
| (b) whooping cough | (ii) inactivated toxin |
| (c) diphtheria | (iii) killed bacteria |
| (d) polio | (iv) harmless bacteria |
| (a) (b) (c) (d) | (i) (ii) (iii) (iv) |
| (1) (iii) (ii) (iv) (i) | (2) (iv) (iii) (ii) (i) |
| (3) (i) (ii) (iv) (iii) | (4) (ii) (i) (iii) (iv) |

166. The name of 'Mary Mallon' is associated with the disease :-

- (1) Typhoid
- (2) Leprosy
- (3) Tuberculosis
- (4) Small pox

167. In full blown AIDS, opportunistic infections are especially caused by :-

- (1) Mycobacterium
- (2) Toxoplasma
- (3) Fungi
- (4) All of these

168. In which one of the following options the two examples are incorrectly matched with their particular type of immunity ?

	Examples	Type of immunity
(1)	Saliva in mouth and tears in eyes	Physiological barriers
(2)	Skin, HCl in stomach and mucus coating	Physical barriers
(3)	Polymorphonuclear leukocytes and monocytes	Cellular barriers
(4)	Anti-tetanus and antsnake bite injection	Passive immunity

169. Maximum concentration of (A) antibody recorded with high sensitive allergens like penicilline and it leads to (B). (A) & (B) respectively are ?

- (1) IgG, Erythroblastosis foetalis
- (2) IgE, Anaphylactic shock
- (3) IgE, Urticaria
- (4) IgM, Rheumatoid arthritis

170. Process or function only shown by IgG & IgM is ?

- (1) Agglutination
- (2) Opsonisation
- (3) Neutralisation
- (4) Phagocytosis

171. First line of defence, second line of defence & third line of defence respectively are ?

- (1) Mucosa, NK cell, Antibodies
- (2) Skin, Fever, Interferones
- (3) Interferone, T-killer cell, Skin
- (4) NK cell, T-killer cell, Antibodies

172. Bluish colouration of lips and finger nails due to infection in alveoli (lower respiratory tract infection) may be caused due to ?

- (1) *Haemophilus Influenzae*
- (2) *Orthomyxo virus*
- (3) *Rhino virus*
- (4) *Corynebacterium diphtheriae*

- | | |
|--|---|
| <p>173. Which statement is true for asymptomatic phase of AIDS :-</p> <p>(1) Tuberculosis caused by mycobacterium avium.</p> <p>(2) Pneumonia caused by pneumocystis carinii.</p> <p>(3) AIDS can be detected during this phase.</p> <p>(4) Many oppurtunistic infections are caused during this phase.</p> <p>174. Which statement is false for 'HAART' :-</p> <p>(1) It contains protease inhibitor only</p> <p>(2) It is only partially effective.</p> <p>(3) It contains protease and reverse transcriptase inhibitor.</p> <p>(4) It is used for the treatment of AIDS.</p> <p>175. Which disease is not caused by bacteria ?</p> <p>(1) Chlamydia (2) Trichomoniasis</p> <p>(3) Gonorrhoea (4) Syphilis</p> <p>176. "Eggs of parasite are excreted along with the faeces of infected person." Given statement is true for which disease ?</p> <p>(1) Amoebiasis (2) Ascariasis</p> <p>(3) Typhoid (4) Shigellosis</p> | <p>177. Bleeding from nose, mouth and gums is seen in which of the following disease ?</p> <p>(1) Filariasis</p> <p>(2) Typhoid fever</p> <p>(3) Dengue fever</p> <p>(4) Shigellosis</p> <p>178. In the given list how many diseases are caused by virus :-
Cholera, Swine flu, Pertussis, Measles, Rabies</p> <p>(1) 3 (2) 5 (3) 2 (4) 4</p> <p>179. Mutation in HBA1 and HBA2 genes on chromosome 16 leads to which disorder ?</p> <p>(1) β-thalassemia</p> <p>(2) Sickle cell anemia</p> <p>(3) α-thalassemia</p> <p>(4) Tay-sach's disease</p> <p>180. Which pair is not matched correctly ?</p> <p>(1) IgG $\Rightarrow$ lightest antibody</p> <p>(2) IgE $\Rightarrow$ Protection from parasites</p> <p>(3) IgM $\Rightarrow$ Oldest antibody</p> <p>(4) IgA $\Rightarrow$ Maximum number of paratopes</p> |
|--|---|